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(54) METHOD FOR PRODUCING A POWER FINFET

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(57) **ABSTRACT**

A method for producing a power FinFET with control electrodes. The power FinFET includes a semiconductor body which includes a second connection region and a drift layer. The second connection region forms a front side of the semiconductor body. The method includes: creating a first structured mask on the front side of the semiconductor body using a first lithography step, wherein the first structured mask includes oxide regions, first open regions and second open regions, wherein the first open regions and the second open regions expose the front side of the semiconductor body; creating first trenches below the first open regions of the mask and second trenches below the second open regions of the mask using a first etching process starting from the front side of the semiconductor body into the drift layer, wherein the first and second trenches trenches are arranged substantially parallel to one another and alternate.

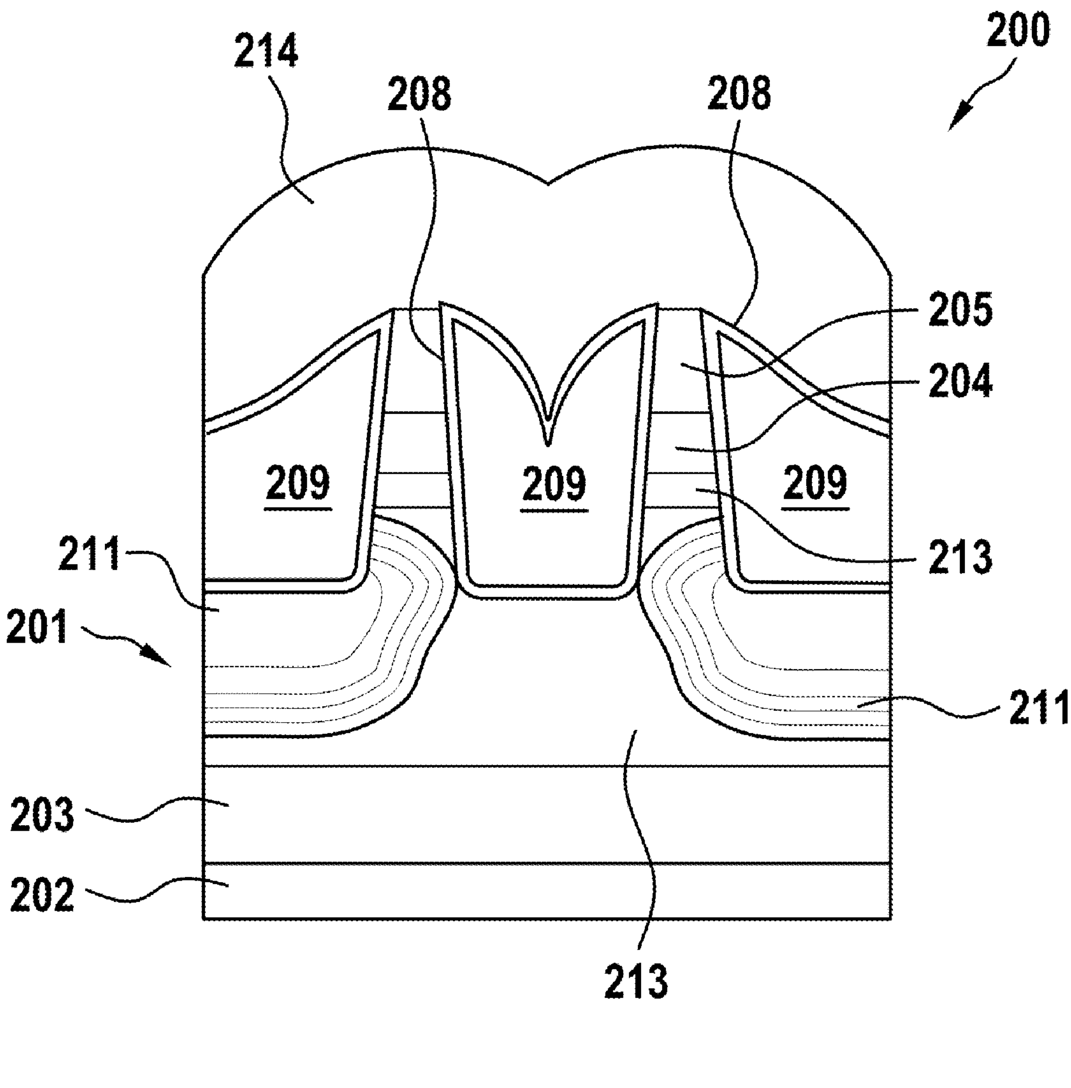


Fig. 1

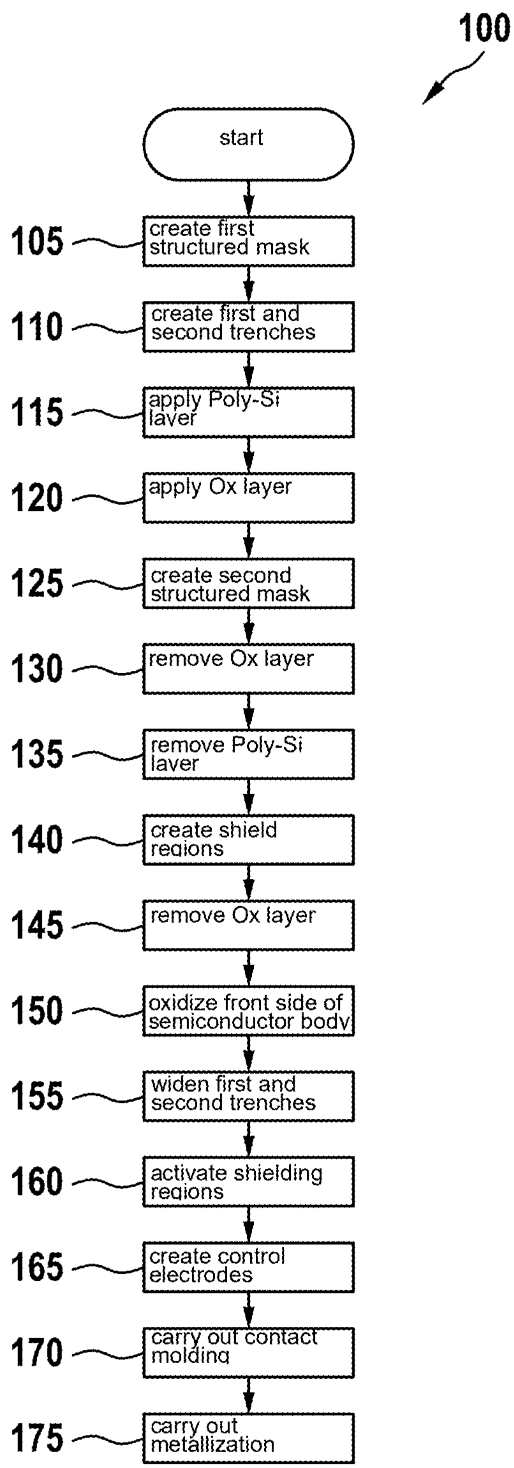
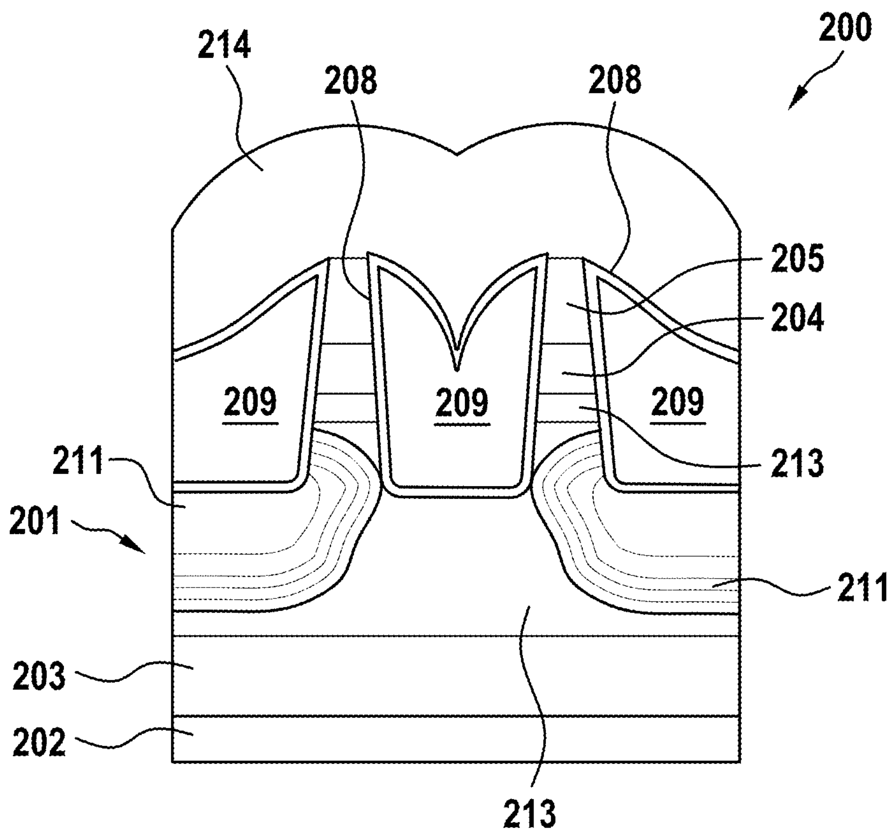

100
start
105
create first structured mask
110
create first and second trenches
115
apply Poly-Si layer
120
apply Ox layer
125
create second structured mask
130
remove Ox layer
135
remove Poly-Si layer
140
create shield regions
145
remove Ox layer
150
oxidize front side of semiconductor body
155
widen first and second trenches
160
activate shielding regions
165
create control electrodes
170
carry out contact molding
175
carry out metallization

Fig. 2



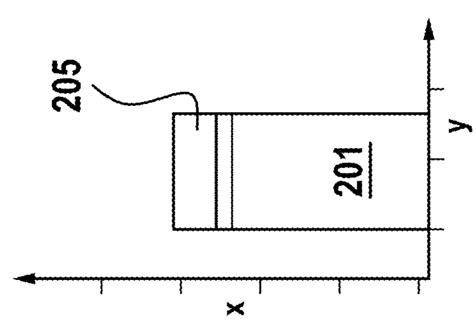


Fig. 3A

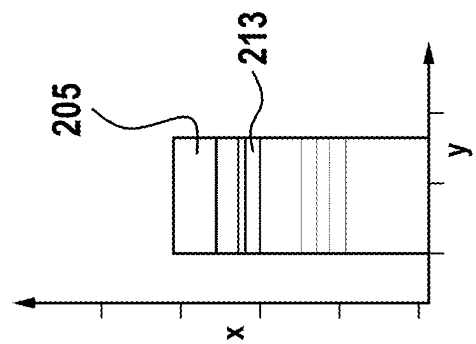


Fig. 3B

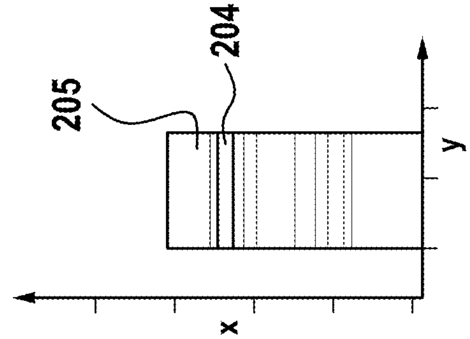


Fig. 3C

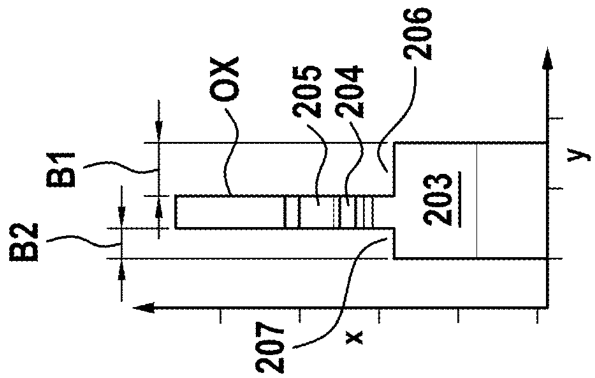


Fig. 3D

Fig. 3G

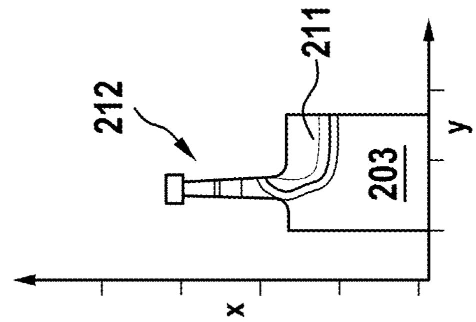


Fig. 3F

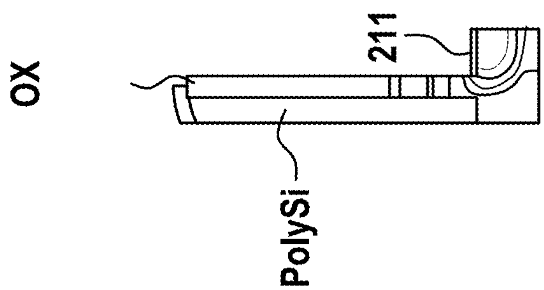


Fig. 3E

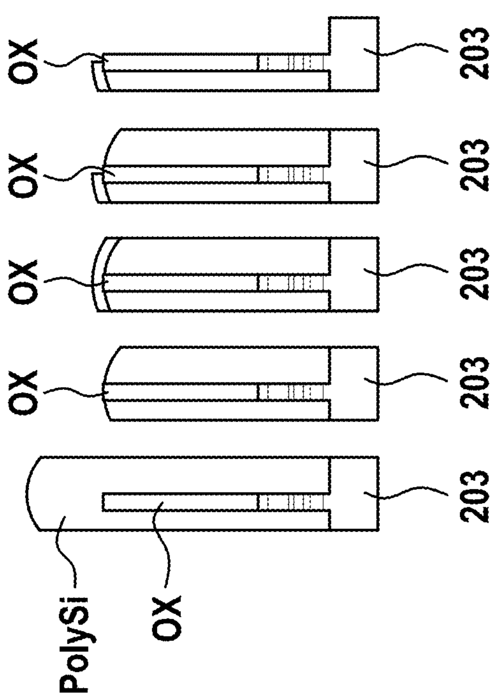


Fig. 3H

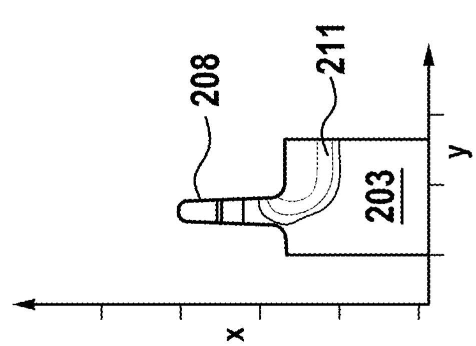


Fig. 3I

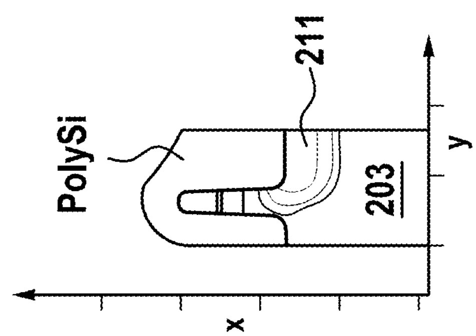


Fig. 3J

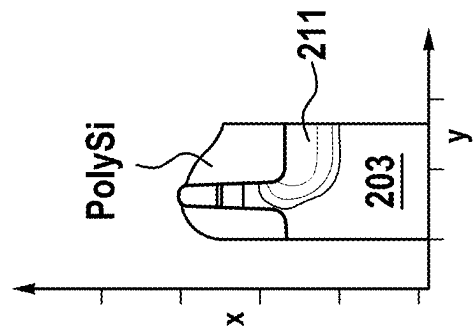


Fig. 3K

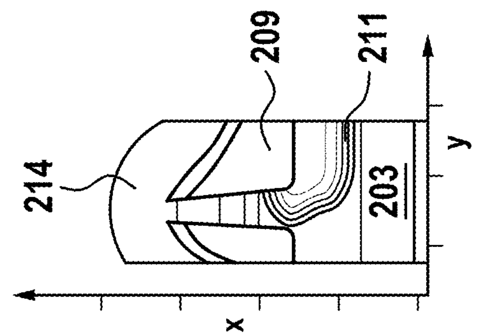
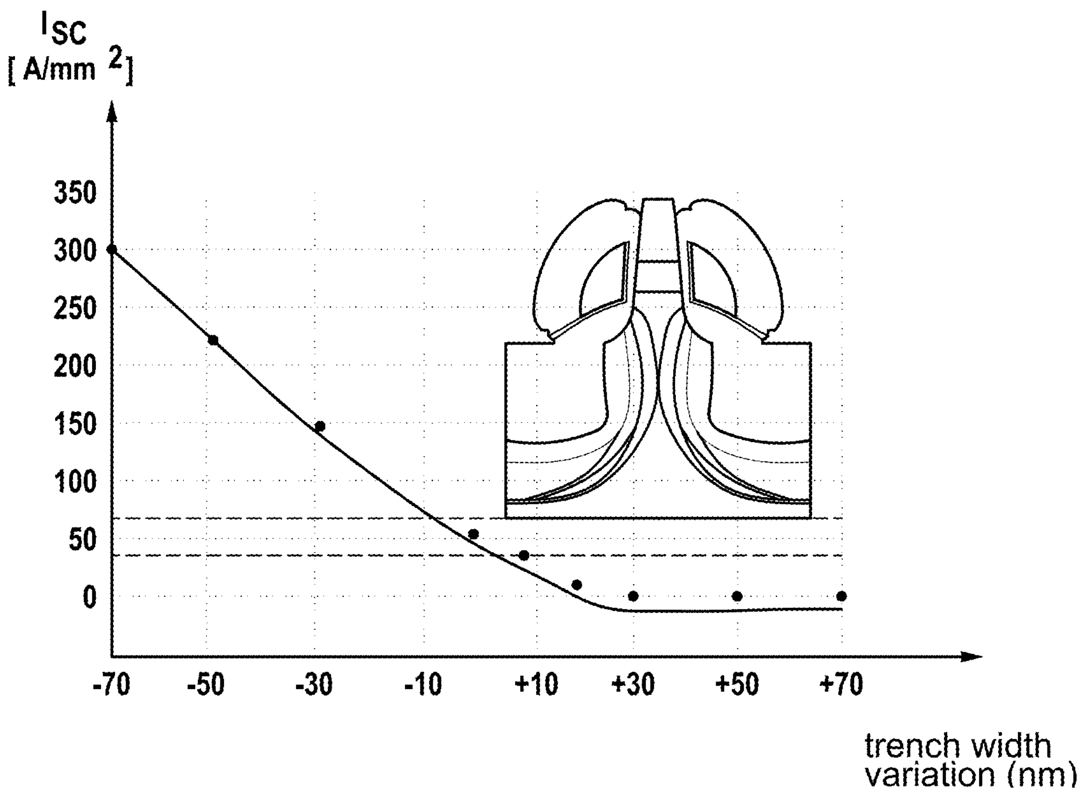


Fig. 4


ISC
[A/mm 2]
350
300
250
200
150
100
50
0
-70
-50
-30
-10
+10
+30
+50
+70
trench width
variation (nm)

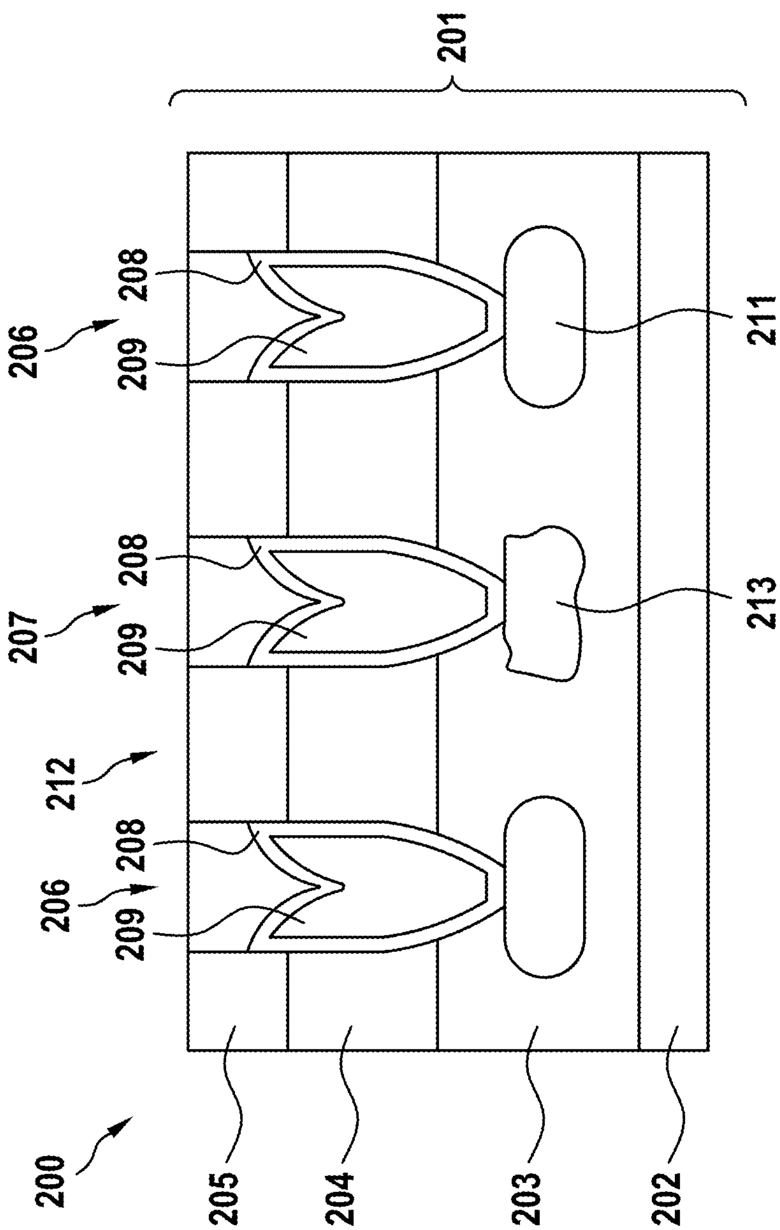


Fig. 5

METHOD FOR PRODUCING A POWER FINFET

FIELD

[0001] The present invention relates to a method for producing a power FinFET by means of lithography masks and to a corresponding power FinFET.

BACKGROUND INFORMATION

[0002] In power electronics, semiconductors with a wide band gap such as SiC or GaN are used. Typically, power MOSFETs with a vertical channel region (so-called TMOSFETs) are used.

[0003] In the TMOSFET design, the n+ source situated in a semiconductor material and the p channel region are interrupted by trenches that can extend up to an n− drift region. Inside the trench, there is a gate electrode, which is separated from the semiconductor material by a gate oxide and is used for controlling the channel region. By suitable selection of the geometry, epitaxial doping, channel doping and screening doping, the on-resistance, threshold value voltage, short-circuit resistance, oxide load and breakdown voltage of the TMOSFET can be optimized.

[0004] To increase the breakdown voltage of such power MOSFETs, shielding regions are arranged below the trenches.

[0005] Since these shielding regions are connected to the source regions, it may be necessary to arrange two-part control electrodes within the trenches, as described in DE 10224201 B4.

[0006] A disadvantage here is that the trenches have to be relatively wide, so the pitch dimension and the on-resistance of the power MOSFET are large.

[0007] Between the shielding regions, which are usually p-doped, a so-called JFET is formed between two adjacent trenches, which serves to limit the current through the channel region in the event of a short circuit. For this purpose, p-doped shielding regions are implanted using a lithographically structured mask.

[0008] A disadvantage here is that due to process fluctuations, the distances between two p-doped shielding regions vary and consequently the limiting of a short-circuit current is unintentionally influenced.

SUMMARY

[0009] The present invention provides, among other things, a production method for producing a power FinFET, and a power FinFET.

[0010] Preferred example embodiments of the present invention are disclosed herein.

[0011] According to a first aspect, the present invention provides a method for producing a power FinFET having control electrodes. According to an example embodiment of the present invention, the power FinFET comprises a semiconductor body which comprises a second connection region and a drift layer, wherein the second connection region forms a front side of the semiconductor body, the method comprising the steps of:

[0012] Creating a first structured mask on the front side of the semiconductor body with the aid of a first lithography step, wherein the first structured mask comprises oxide regions, first open regions and second open regions, wherein the first open regions and the second open regions expose the front side of the semiconductor body,

[0013] Creating first trenches below the first open regions of the mask and second trenches below the second open regions of the mask with the aid of a first etching process starting from the front side of the semiconductor body into the drift layer, wherein the first trenches and the second trenches are arranged substantially parallel to one another and alternate, wherein the second trenches comprise substantially the same width as the first trenches, applying a polysilicon layer to the front side of the semiconductor body so that the first trenches and the second trenches are filled, applying an isotropic oxide layer to the front side of the semiconductor body,

[0014] Creating a second structured mask on the isotropic oxide layer using a second lithography step, wherein the second structured mask is open above the first trenches, removing the isotropic oxide layer above the first trenches with the aid of a second etching process, removing the polysilicon layer within the first trenches with the aid of a third etching process, creating shielding regions below the first trenches with the aid of a first implantation process, removing the isotropic oxide layer above the second trenches and the polysilicon layer within the second trenches with the aid of a fourth etching process, oxidizing the front side so that a further oxide layer is arranged on the front side, widening the first trenches and the second trenches with the aid of a fifth etching process so that fins are formed between the first trenches and the second trenches, wherein the fins preferably comprise a width of less than 500 nm, activating the shielding regions by means of annealing, and creating control electrodes within the first trenches and within the second trenches.

[0015] In one possible embodiment of the production method of the present invention, in order to create the control electrodes an electrode material for the control electrodes, in particular polysilicon, is deposited in a layer thickness in such a way that the trenches are completely filled, wherein after a subsequent etching process the trenches remain completely filled with the electrode material for the control electrodes.

[0016] The contact to the p-shielding regions below every second trench is therefore not made by a contact in the trench, but by a contact at the end of the cell field, or by a deep p-terminal region implanted at periodic intervals along the fin.

[0017] An advantage of the production method of the present invention is that a short-circuit current-limiting effect occurs between the shielding region and the side walls of the second trenches. As a result, process fluctuations can be tolerated. Although a second lithography mask is used to open the first trenches, the position of the shielding implantation is not subject to any adjustment tolerance since the position of the shielding region is defined by the trenches themselves.

[0018] In a development of the present invention, the first structured mask has nitride regions, wherein the oxide regions are arranged on the nitride regions.

[0019] The advantage here is that oxidation of the fin top is prevented.

[0020] In a further embodiment of the present invention, spreading regions below the second trenches are created with the aid of a second implantation process, wherein the second implantation energy comprises a value between 60 keV and 2500 keV.

[0021] The advantage here is that the on-resistance of the power FinFET is reduced.

[0022] In a development of the present invention, the first etching process and the second etching process are anisotropic plasma etching processes.

[0023] An advantage here is that the structured masks can be transferred to the underlying layers with minimal widening.

[0024] In a possible embodiment of the production method of the present invention, the first implantation process comprises a first implantation energy in the range of 30 keV to 2700 keV.

[0025] The advantage here is that the shielding regions are produced in the trench bottom below the gate oxide to be protected, so that a maximum shielding effect is achieved without pitch loss.

[0026] According to a further aspect, an example embodiment of the present invention provides a power FinFET having control electrodes and a semiconductor body that comprises a drift layer and a second connection region, wherein the second connection region is arranged above the drift layer and wherein first trenches and second trenches extend from the second connection region into the drift layer of the semiconductor body, wherein the first trenches and the second trenches are arranged alternating relative to one another, wherein the first and second trenches typically comprise the same width, wherein shielding regions are arranged below the first trenches, wherein the shielding regions are immediately adjacent to the first trenches,

[0027] wherein a control electrode is arranged within each of the first trenches and a control electrode is arranged within each of the second trenches,

[0028] wherein the control electrode arranged in a first trench is electrically insulated from a shielding region located below the first trench, and wherein fins are arranged between the first trenches and the second trenches, wherein the fins preferably comprise a width of at most 500 nm.

[0029] An advantage here is that the short-circuit current is limited by the space-charge zone of the shielding regions and of the opposite trench wall of a second trench. Furthermore, it is advantageous that the influence of process variability on the short-circuit current and the on-resistance of the power FinFET is reduced.

[0030] In a development of the present invention, spreading regions are arranged below the second trenches.

[0031] The advantage here is that the current propagation is high and the on-resistance is low.

[0032] In a further embodiment of the present invention, the shielding regions are p-doped and have a dopant concentration of at least $1E18/cm^3$.

[0033] This is advantageous in that high implantation doses can be introduced cost-effectively below the trench bottom and deeper regions can be created with low implantation energies.

[0034] In one example embodiment of the present invention, the semiconductor body of the power FinFET comprises silicon carbide (SiC).

[0035] The advantage here is that aluminum, which is easily activated, can be used for implantation.

[0036] In another embodiment of the present invention, the semiconductor body of the power FinFET comprises gallium nitride (GaN).

[0037] The advantage here is that the critical field strength and the electron mobility are high.

[0038] In one embodiment of the power FinFET of the present invention, the control electrodes located in the first trenches are formed in one part.

[0039] Further advantages of the production method according to the present invention and of the power FinFET according to the present invention result from the following description of embodiments.

BRIEF DESCRIPTION OF THE DRAWINGS

[0040] The present invention is explained below with reference to preferred embodiments and the figures.

[0041] FIG. **1** shows a method for producing a power FinFET with a one-part control electrode, according to an example embodiment of the present invention.

[0042] FIG. **2** shows a first embodiment of a power FinFET with a one-part control electrode, according to an example embodiment of the present invention.

[0043] FIG. **3**A to **3**K are sectional views showing production steps of the production method according to an example embodiment of the present invention.

[0044] FIG. **4** is a diagram for explaining the operation of a power FinFET according to an example embodiment of the present invention.

[0045] FIG. **5** shows a second example embodiment of a power FinFET with a one-part control electrode, according to the present invention.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

[0046] FIG. **1** shows a flow chart of a possible embodiment of a method according to the present invention for producing a power FinFET **200** with control electrodes. A power FinFET **200** with control electrodes **209** produced using the method is shown in FIG. **2**. FIGS. **3**A-**3**K show sectional views for explaining individual production steps.

[0047] The power FinFET **200** comprises a semiconductor body **201** which comprises for example silicon carbide (SiC) or gallium nitride (GaN). The power FinFET **200** comprises a first connection region **202** on the back side (bottom in FIG. **2**), a drift layer **203**, a channel region **204** and a second connection region **205** on the front side (top in FIG. **2**).

[0048] For producing the power FinFET **200**, the semiconductor wafer or semiconductor body **201** having the corresponding n-drift epitaxial layer **203** is first provided in preparatory steps **100** (see also FIG. **3**A). This is followed by an n-source implantation for the second connection region **205** (FIG. **3**B) and an n spreading implantation for spreading regions **213** to be produced (FIG. **3**C).

[0049] In a step **105**, a first structured mask M**1** is created on the front side of the semiconductor body **201** with the aid of a first lithography step (FIG. **3**D). The first structured mask M**1** comprises oxide regions Ox, first open regions B**1** and second open regions B**2**, wherein the first open regions B**1** and the second open regions B**2** of the first structured mask expose the front side of the semiconductor body **201**.

[0050] In a subsequent step **110**, first trenches **206** below the first open regions B**1** of the mask M**1** and second trenches **207** below the second open regions B**2** of the mask M**1** are created with the aid of a first etching process starting from the front side of the semiconductor body **201** into the drift layer **203** of the semiconductor body **201** (see FIG. **3**D). In other words, the first trenches **206** and the second trenches **207** are preferably created simultaneously. The first trenches **206** and the second trenches **207** are arranged substantially parallel to one another and alternate. In a preferred embodiment, the width B**2** of the second trenches **207** is typically equal to the width B**1** of the first trenches **206**. The first etching process performed in step **110** is preferably an anisotropic etching process. In the case of a SiC semiconductor body **201**, the first etching process selects between silicon carbide (SiC), which is etched, and silicon dioxide (SiO2), silicon nitride (SiN) and silicon (Si), which are etched as little as possible.

[0051] In a subsequent step **115**, a polysilicon layer Poly-Si is applied onto the front side of the semiconductor body **201** so that the first trenches **206** and second trenches **207** are filled (see also FIG. **3**E at left).

[0052] In a subsequent step **120**, an isotropic oxide layer Ox is applied to the front side of the semiconductor body **201**.

[0053] In a subsequent step **125**, a second structured mask M**2** is created on the isotropic oxide layer with the aid of a second lithography step, wherein the second structured mask M**2** is open above the first trenches **206** (see FIG. **3**E).

[0054] In a subsequent step **130**, the isotropic oxide layer Ox above the first trenches **206** is removed with the aid of a second etching process (see FIG. **3**E). The second etching process performed in step **130** is preferably an anisotropic etching process. The second etching process etches the oxide layer, in particular silicon dioxide (SiO2), whereas silicon (Si) is etched as little as possible

[0055] In a subsequent step **135**, the polysilicon layer Poly-Si within the first trenches **206** is removed with the aid of a third etching process (see FIG. **3**E at the right). The third etching process performed in step **135** removes silicon (Si). However, the third etching process is very selective to silicon dioxide (SiO2), silicon nitride (SiN) and silicon carbide (SiC), which are not etched.

[0056] In a subsequent step **140**, shielding regions **211** are created below the first trenches **206** with the aid of a first implantation process (FIG. **3**F). The first implantation energy is between 30 keV and 2700 keV. The created shielding regions **211** are p-doped.

[0057] In a subsequent step **145**, the isotropic oxide layer Ox above the second trenches **207** and the polysilicon layer Poly-Si within the second trenches **207** are removed with the aid of a fourth etching process (FIG. **3**G). The fourth etching process performed in step **145** etches the oxide layer, particularly silicon dioxide (SiO2), and silicon (Si), but is selective to silicon nitride (SiN) and silicon carbide (SiC), which are not etched.

[0058] In a subsequent step **150**, the front side of the semiconductor body **201** is oxidized so that a further oxide layer is arranged on the front side of the semiconductor body **201**. The oxide layer comprises a thickness of at least 10 nm.

[0059] In a subsequent step **155**, the first trenches **206** and the second trenches **207** are widened with the aid of a fifth etching process so that fins **212** are formed between the first trenches **206** and the second trenches **207**. The fins **212** formed preferably comprise a width of less than 500 nm (FIG. **3**G). The oxide from step **150** is selectively etched wet-chemically. The fifth etching process in the case of a SiC semiconductor body selects between oxide, in particular silicon dioxide (SiO2), which is etched, and silicon carbide (SiC) and silicon nitride (SiN), which are not etched.

[0060] The steps **150** and **155** are carried out cyclically depending on the fin width of the fins **212** to be achieved. In other words, the front side of the semiconductor body **201** is oxidized multiple times, wherein an etching step is carried out between the oxidation steps. The widening of the trenches **206**, **207** (see FIG. **3**G) is thus carried out without adjustment, or in a self-adjusting manner. The lateral oxidation rate exceeds the vertical oxidation rate by approximately a factor of two.

[0061] In a step **160**, the shielding regions **211**, which lie below the first trenches **206**, can then be activated by means of annealing. The annealing is typically carried out at about 1700° C. Furthermore, a deposition of a gate oxide **208** can be carried out (FIG. **3**H).

[0062] In a subsequent step **165**, the control electrodes **209** are created within the first trenches **206** and within the second trenches **207**. For this purpose, the electrode material for the control electrodes **209**, in particular polysilicon, is first deposited in a layer thickness so that all trenches **206**, **207** (i.e., both the first trenches **206** and the second trenches **207**) are completely filled (see FIG. **3**I). In a subsequent sixth etching process in step **165**, this has the result that not only does a spacer remain on the trench side walls of the trenches **206**, **207**; rather, all trenches **206**, **207** remain completely filled with the electrode material for the control electrode **209**, in particular with polysilicon Poly-Si (see FIG. **3**J). The contact to the p-shielding regions **211**, which are arranged below every second trench (i.e., below the first trenches **206**), is therefore not made by a contact in the trench, but by a contact at the end of the cell field, or by a deep p-connection region implanted at periodic intervals along the fin **212**. Furthermore, in step **165** a gate polysilicon insulation is carried out. This comprises an oxide deposition and/or poly-reoxidation.

[0063] In a further step **170**, contact molding is carried out at the bottom of the trench, at the fin tip of the fin **212** and at the rear side.

[0064] Finally, in a further step **175**, a metallization **214** of the front and back sides is carried out (see also FIG. **3**K).

[0065] FIG. **2** shows a possible embodiment of a power FinMOSFET **200** produced using the method according to the present invention. After the fins **212** have been molded, the first trenches **206** are widened to such an extent that there is sufficient space for the gate or control electrode **209**, the gate and insulation oxides **208** and the p-doped shielding regions **211**. The second trenches **207** created between the first trenches **206** are made with the same width as the first trenches **206**, so that in the poly-recess process in the step **170** the trench floor or trench bottom is not exposed either in the first trenches **206** or in the second trenches **207**. The n-region below the second trench **207** thus remains isolated from the gate or control electrode **209** and the source contact. The under-scattering of the p-shield implantation into the first trenches **206** together with the width of the second trenches **207** and the resulting width of the fins **212** controls the distance of the p-shielding region **211** from the opposite trench edge. With these design parameters, an optimum of on-resistance Ron, short-circuit current Isc and field strength in the oxide at the trench bottom of the second trench **207** can be set. The p-channel region can be contacted in the third dimension via p+ doped regions that alternate with the n-source regions along the fin **212**.

[0066] The power FinFET **200** shown in FIG. **2** has a semiconductor body **201** that comprises a first connection region **202**, a drift layer **203**, a channel region **204** and a second connection region **205**.

[0067] The first connection region **202** of the semiconductor body **201** preferably functions as a drain connection and the second connection region **205** of the semiconductor body **201** preferably functions as a source connection. The drift layer **203** of the semiconductor body **201** is arranged on the first (rear) connection region **202** (bottom in FIG. **2**). The channel region **204** of the semiconductor body **201** is arranged on the drift layer **203**. The second connection region **205** of the semiconductor body **201** or the source connection **205** is arranged on the channel region **204** of the semiconductor body **201**. The second connection region or source connection **205** is located on the front side of the semiconductor body **201** (top in FIG. **2**). The semiconductor body **201** preferably comprises silicon carbide (SiC) or gallium nitride (GaN).

[0068] Starting from the front side of the semiconductor body **201**, first trenches **206** and second trenches **207** preferably extend into the drift layer **203** of the semiconductor body **201**, wherein the second trenches **207** typically comprise the same width as the first trenches **206**. The first trenches **206** and the second trenches **207** are arranged in an alternating manner. Shielding regions **211**, which are preferably p-doped, are arranged below the first trenches **206**. The shielding regions **211** directly adjoin a trench bottom of the first trenches **206**. The dopant concentration of the shielding regions **211** is at least $1E18/cm^3$.

[0069] A control electrode **209**, which acts as a gate connection, is arranged within each first trench **206**. The control electrode **209** is electrically insulated from the shielding region **211** with the aid of an oxide layer **208**. Between the first trenches **206** and the second trenches **207**, fins **212** are arranged which preferably comprise a width of less than 500 nm.

[0070] With the aid of the production method according to the present invention, it is achieved that the shielding regions **211** below the first trenches **206** are further apart from one another than the shielding regions **211** are from the opposite trench walls or side walls of the second trenches **207**. As a result, the short-circuit current Isc is not limited by the collision of the space-charge zones of two shielding regions, but by the space-charge zone of each p-doped shielding region **211**, which displaces or pushes the current against the opposite trench wall of a second trench **207**. The low sensitivity to process variability is achieved by the fact that, in the event of a short circuit, the trench wall of the second trench **207** forms an accumulation channel due to the positive gate voltage, which accumulation channel cannot be cleared by the space-charge zone of the p-doped shielding region **211**.

[0071] FIG. **4** shows the consequences for a short-circuit current Isc when the p-shielding regions have different distances relative to one another using the example of an implantation through a trench in which the trench width is varied. As soon as the p-shielding regions are only 10 nm too far apart (10 nm narrower trench), the short-circuit current Isc is no longer sufficiently limited and is above the tolerance limit. Likewise, a trench that is 30 nm wider (or an implantation mask opening that is 30 nm wider) results in the neighboring p-regions touching one another, the JFET is completely closed and thus no current flows at all and the on-resistance of the power FinFET consequently approaches infinity.

[0072] In the power FinFET **200** according to the present invention with alternating p-shielding regions **211**, this is prevented by implanting only in every second trench, i.e. in the first trenches **206**. As a result, the effect limiting the short-circuit current Isc between a p-shielding region **211** and a trench edge is created, which is less sensitive to process variations.

[0073] In one possible embodiment, the p-shielding regions **211** are implanted by implantation with a lithographically structured mask.

[0074] Alternatively, the implantation can also be carried out through a trench. In both cases, the distance between two adjacent p-shielding regions **211** is subject to a certain process fluctuation.

[0075] A further advantage is that below a second trench **207** (below which there is no p-shielding region) the distance between two adjacent p-shielding regions **211** is increased, thus creating a space in which the electric current I can propagate. This spreading region **213** results in the on-resistance of the power FinFET **200** being reduced.

[0076] In addition, it is possible to n-dope the spreading region **213** slightly higher than the n-drift zone **203** in order to further increase the current propagation effect. This can be realized by means of implanting into the non-p-implanted trenches. As a result, the implantation energies required to produce the spreading region **213** are not as high as those required for an initial implantation through the unstructured wafer.

[0077] In an exemplary embodiment, the first structured mask M1 comprises nitride regions (in particular SiN), which are located between the front side and the oxide regions. The nitride regions protect the front side or surface of the fins **212**, since oxidation of the fin top is prevented in this way in step **150**. The nitride regions are removed in an intermediate step (not shown in the flow diagram in FIG. **1**) between step **155** and step **160**.

[0078] In a possible further embodiment, in the preparatory steps **100**, the spreading regions **213** can be implanted below the second trenches **207** with the aid of an implantation process. The spreading regions **213** are n-doped and comprise a higher doping than the n-doped drift layer **203**. As a result, the current propagation effect below the second trenches **207** is enhanced. The implantation process preferably comprises an implantation energy that comprises a value in a range between 60 keV and 2500 keV.

[0079] The power FinFET **200** is used primarily in DC/DC converters and inverters of an electric drive train of electric or hybrid vehicles, as well as in vehicle chargers.

1-13. (canceled)

14. A method for producing a power FinFET having control electrodes, wherein the power FinFET includes a semiconductor body which includes a second connection region and a drift layer, wherein the second connection region forms a front side of the semiconductor body, the method comprising the following steps:

creating a first structured mask on the front side of the semiconductor body using a first lithography step, wherein the first structured mask includes oxide regions, first open regions, and second open regions, wherein the first open regions and the second open regions expose the front side of the semiconductor body;

creating first trenches below the first open regions of the first structured mask and second trenches below the second open regions of the first structured mask using a first etching process starting from the front side of the semiconductor body into drift layer, wherein the first trenches and the second trenches are arranged substantially parallel to one another and alternate, wherein the second trenches are substantially the same width as the first trenches;

applying a polysilicon layer onto the front side of the semiconductor body so that the first trenches and second trenches are filled;

applying an isotropic oxide layer to the front side of the semiconductor body;

creating a second structured mask on the isotropic oxide layer using a second lithography step, wherein the second structured mask is open above the first trenches;

removing the isotropic oxide layer above the first trenches using a second etching process;

removing the polysilicon layer within the first trenches using a third etching process;

creating shielding regions below the first trenches using a first implantation process;

removing the isotropic oxide layer above the second trenches and the polysilicon layer within the second trenches using a fourth etching process;

oxidizing the front side so that a further oxide layer is arranged on the front side;

widening the first trenches and the second trenches using a fifth etching process so that fins are formed between the first trenches and the second trenches, wherein the fins have a width of less than 500 nm;

activating the shielding regions by annealing; and

creating control electrodes within the first trenches and within the second trenches.

15. The method according to claim **14**, wherein, for creating the control electrodes, an electrode material for the control electrodes, including polysilicon, is deposited in a layer thickness in such a way that the first and second trenches are completely filled, wherein after a subsequent etching process the first and second trenches remain completely filled with the electrode material for the control electrode.

16. The method according to claim **14**, wherein the first structured mask includes nitride regions, wherein the oxide regions are arranged on the nitride regions.

17. The method according to claim **14**, wherein spreading regions are created below the second trenches using a second implantation process that includes a second implantation energy in a range between 60 keV and 2500 keV.

18. The method according to claim **14**, wherein the first etching and the second etching process are anisotropic plasma etching processes.

19. The method according to claim **14**, wherein the first implantation process includes a first implantation energy in a range of 30 keV to 2700 keV.

20. The method according to claim **14**, wherein the control electrodes formed in the first trenches are formed in one piece.

21. A power FinFET, comprising:

a semiconductor body that includes includes a drift layer and a second connection region;

wherein the second connection region is arranged above the drift layer, and first trenches and second trenches extend from the second connection region into the drift layer of the semiconductor body;

wherein the first trenches and the second trenches are arranged alternating relative to one another and the second trenches have substantially the same width as the first trenches;

wherein control electrodes are arranged within the first trenches, and control electrodes are arranged within the second trenches;

wherein each control electrode arranged in a first trench is electrically insulated from a shielding region located below the first trench; and

wherein between the first trenches and the second trenches, fins are arranged which have a width of at most 500 nm.

22. The power FinFET according to claim **21**, wherein spreading regions are arranged below the second trenches.

23. The power FinFET according to claim **21**, wherein the shielding regions arranged below the first trenches are p-doped and include a dopant concentration of at least $1E18/cm^3$.

24. The power FinFET according to claim **21**, wherein the semiconductor body includes silicon carbide (SiC).

25. The power FinFET according to claim **21**, wherein the semiconductor body includes gallium nitride (GaN).

26. The power FinFET according to claim **21**, wherein the control electrodes formed in the first trenches are formed in one piece.

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